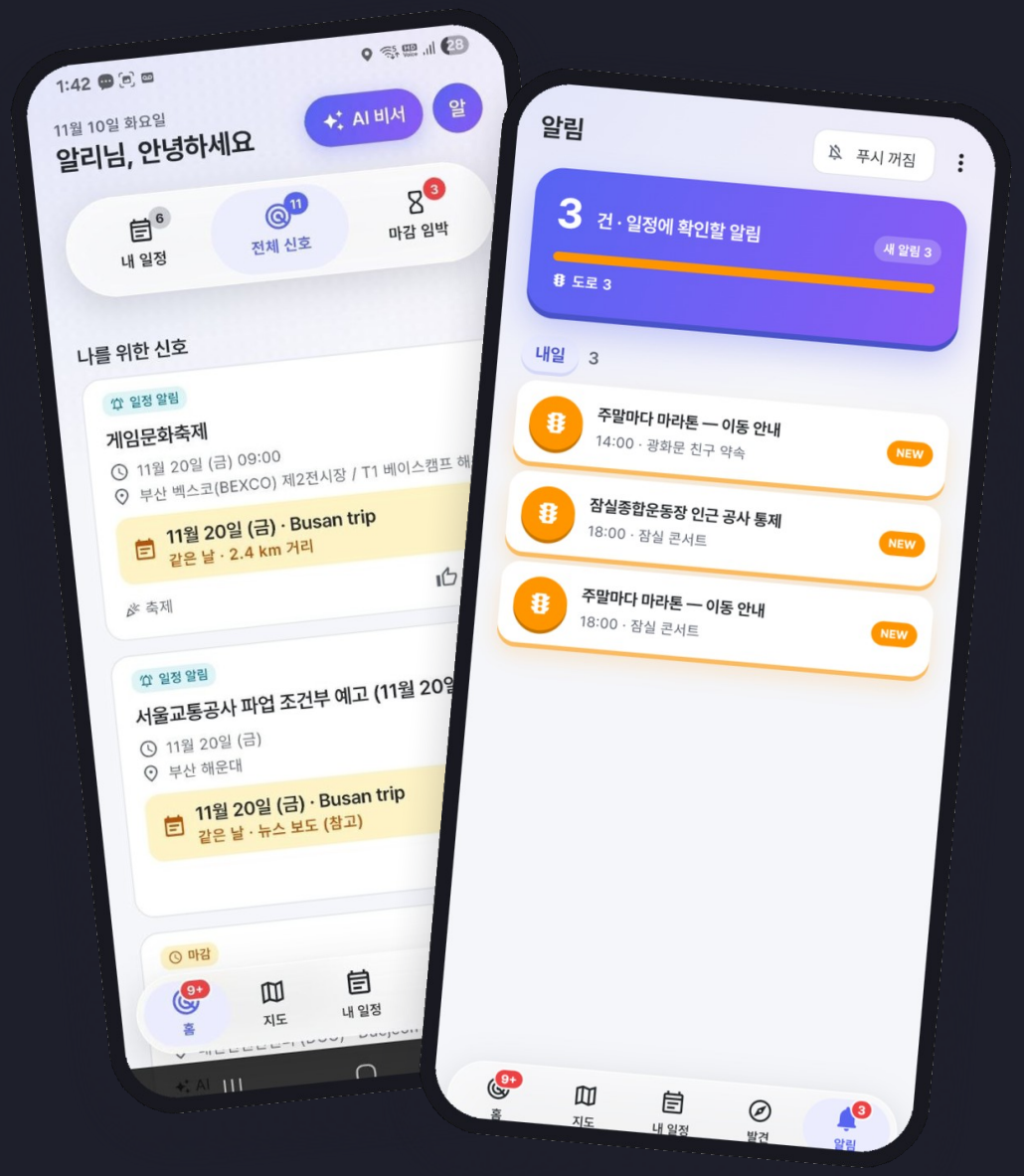
LifeRadar

The AI coworker that watches your schedule
— so you never miss what matters to you.

Finds first **Explains why** **You decide**



2026 AI Woowangthon · Team 27 DALAB · Abbos Aliboev · Jeon Yeongwoo

“Can AI be a coworker?”

Yes — if it finds things before you ask, shows its reasons and sources, and never decides for you.



Finds first

Scans your calendar against real events, official traffic notices, news and weather.



Shows why

Every card says which plan it touches, how far, and where the fact came from.



You decide

Add to calendar, apply, ignore — nothing happens until you tap.

It started with an event we found too late

“I only found the event after the application deadline had passed.”

Our teammate, an international student in Korea. The notice was in Korean, on one of dozens of agency sites — and nothing connected it to his calendar.

Scattered and Korean-only

Events, grants, traffic notices and strike news live on separate sites, mostly in Korean.

Not linked to my plans

A festival 2 km from my hotel on my trip day? I would have to check three apps to notice.

You must know what to ask

Search, maps and chatbots answer questions — they never tell you first.

What matters lives where time, place and information meet



Time

Calendar apps know when you are busy

≠



Place

Map apps know where things are

≠



Information

Event sites, news and traffic notices know what is happening

The useful part is the overlap

…and nobody computes it for you.



Example: a Busan trip on Nov 20

Rail unions announce a strike for that date · a big festival runs 2.4 km from the hotel · a seminar sits right next to the KTX route — three facts, three different sites, one plan.

Today, every tool waits for you to ask

	Calendar app	Map app	Event platforms	AI chatbot	LifeRadar
Tells you before you ask	◐	—	◐	—	●
Links your plans × events × traffic	—	—	—	◐	●
Opportunities along your route (+km, +min)	—	◐	—	—	●
Tracks application deadlines	—	—	●	—	●
Reason + source + confidence	—	—	—	◐	●
Blocks made-up facts	—	—	—	◐	●

● strong ◐ partial — none Team comparison of typical products; features change over time.

LifeRadar crosses them for you — and tells you why

Your plans
Google Calendar + your own

Your interests
chosen, or suggested by AI

658 real events
+ public contests, daily

Official traffic
Seoul TOPIS · national ITS

Strike news
read by AI, quote-checked

Weather
10-day forecast per place

→

Rule engine
Decides what to show: same day? within 1 km? on your route? deadline soon?

Gemini
Writes the “why it matters to you” sentence from those facts only

Code check
Rejects any number, id or quote that is not in the input

→

You decide

- Add to calendar
- Apply before the deadline
- Leave earlier / change route
- Like / dislike — it learns your taste

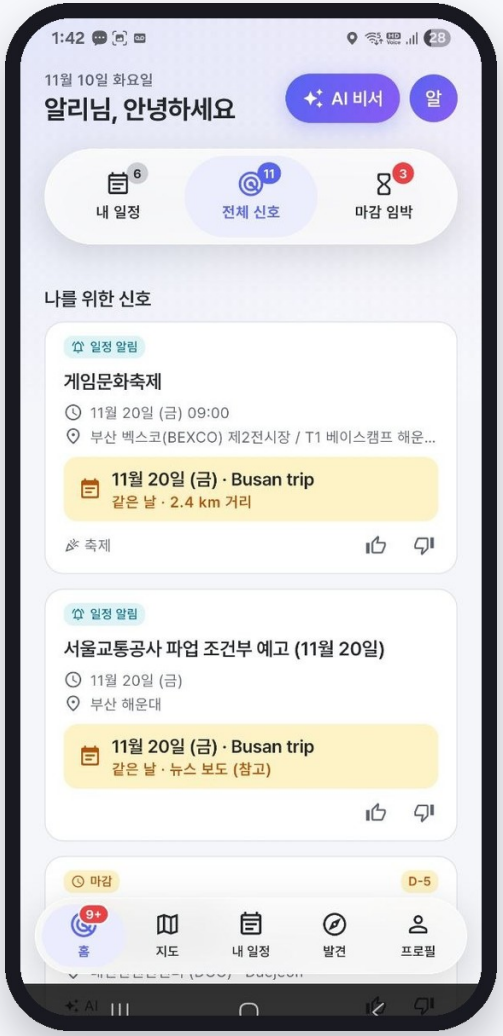
One trip, four signals you didn’t ask for

Rail strike announced

News read by AI, shown with the article quote and link — marked as news, not official.

Festival 2.4 km from your hotel

Real TourAPI event on your trip day. Traffic impact not confirmed — and we say so.



Calendar: “Busan trip · Nov 20–22”

Seminar on your KTX route

+7.1 km · +17 min detour. By car it is not suggested — the highway is too far.

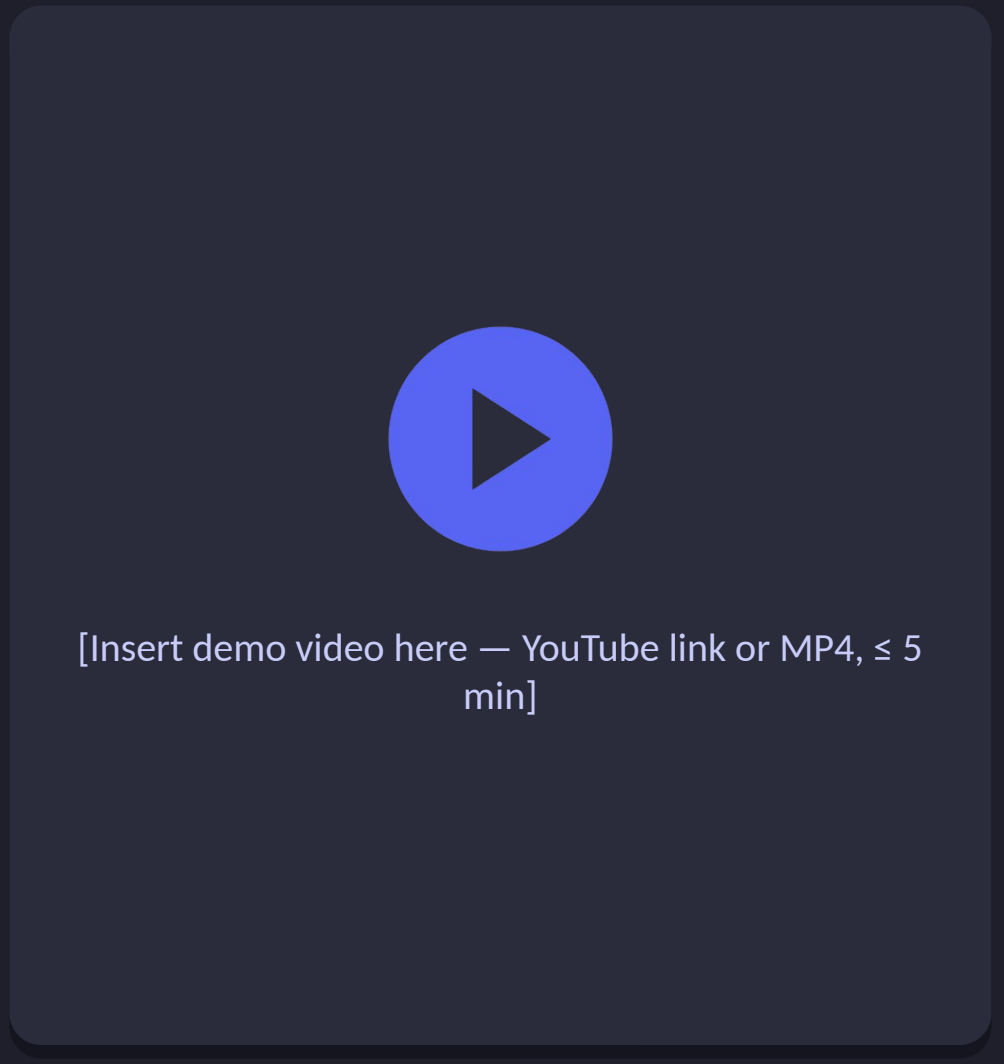
Contest deadline tomorrow

A public contest matching your interests → apply link, or add the deadline to your calendar.

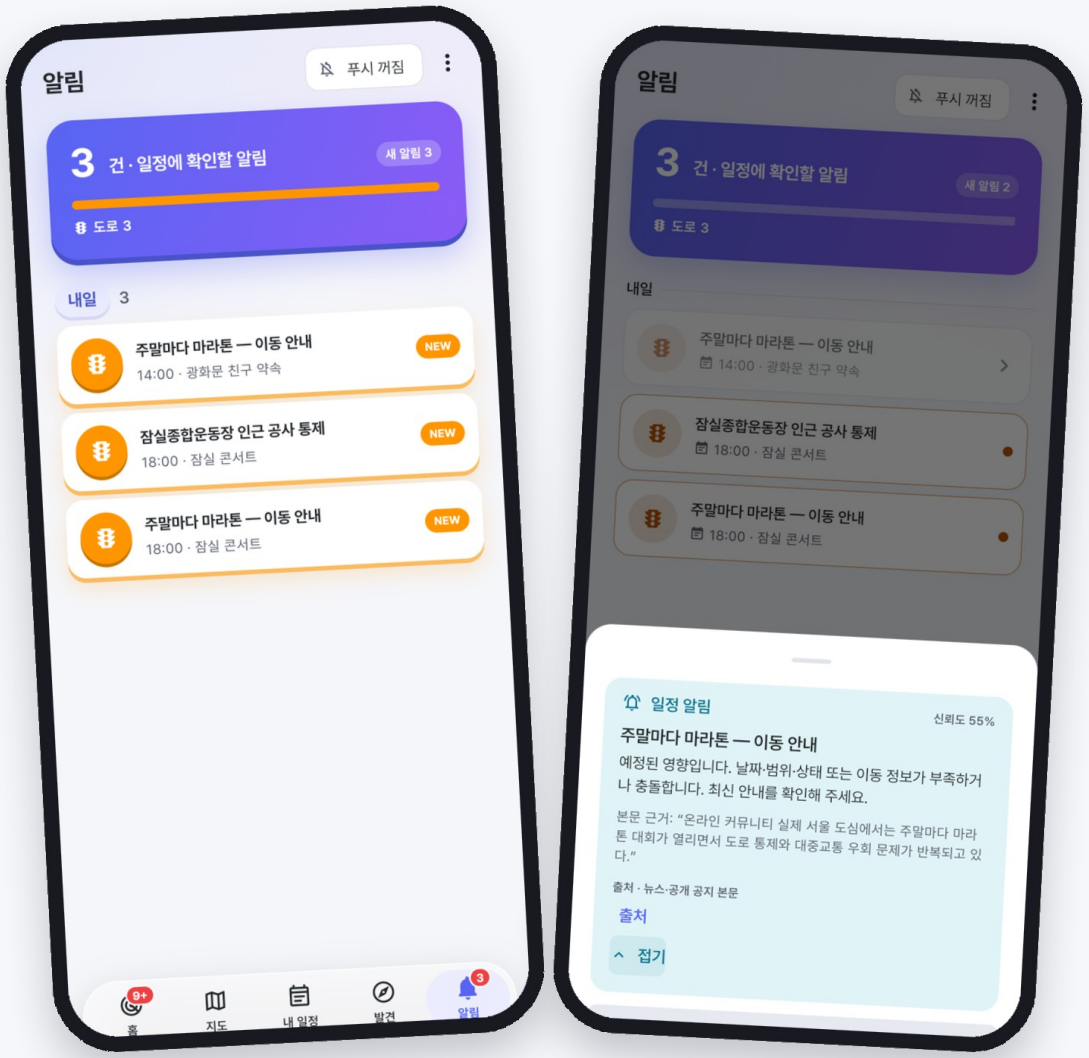
Live demo

Real phone · real data · nothing pre-recorded as “fake results”.

1. Onboarding: tell the AI about yourself
2. Alerts tab: the Busan trip signals
3. Plans: “+ Add route” between two plans
4. AI assistant: [Add] one, [Cancel] one
5. Add a 2-month exhibition → pick one day



Alerts that only talk about your plans



Road

Official closures within 1 km, in your travel window

Transit

Strikes on the line you actually take

Weather

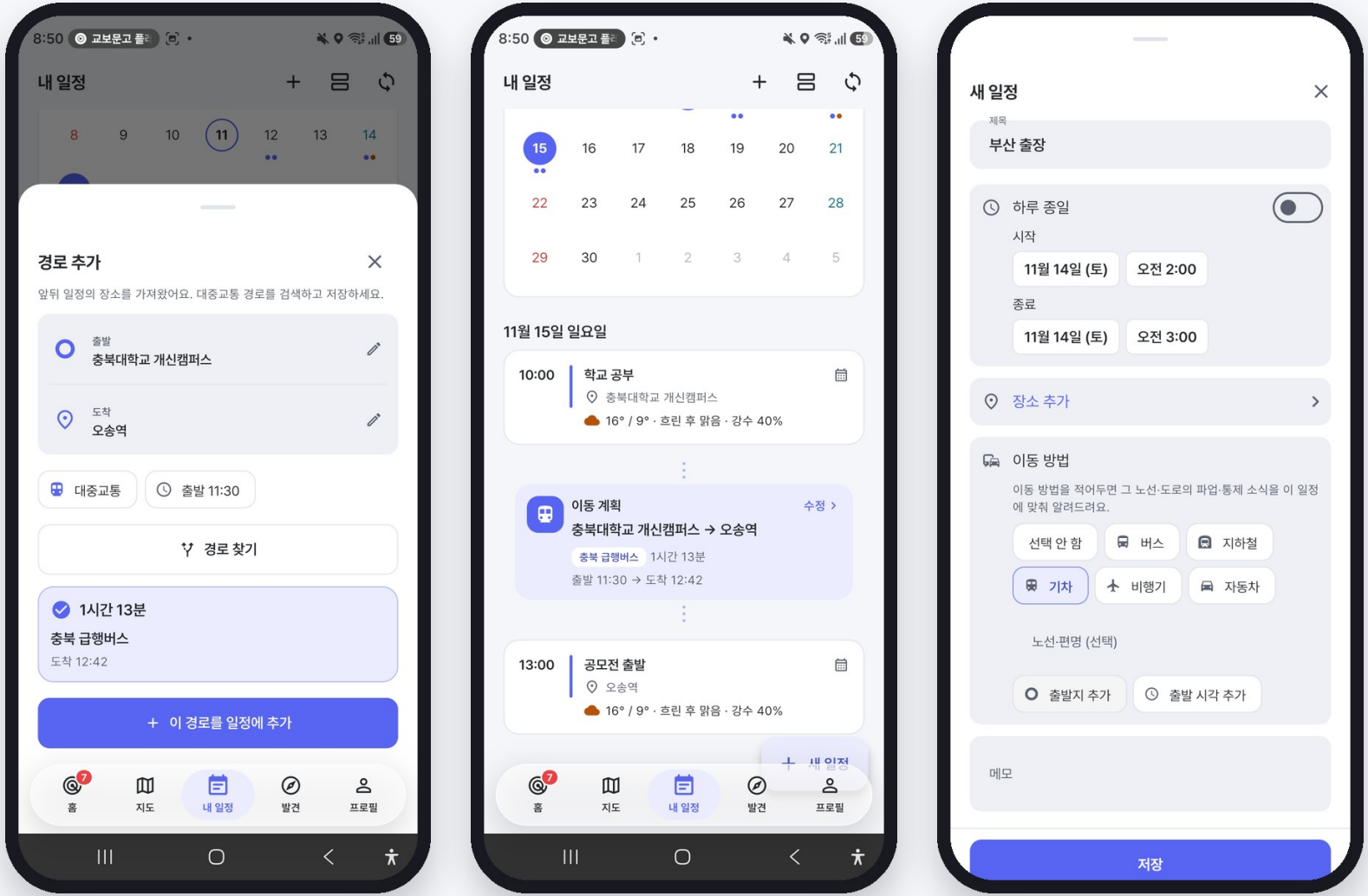
Rain on the day of an outdoor plan

Crowds

Big events next to your plan — “reference”, not a claim

Two lines first · “Read more” for evidence and source · one tap for push

Tell it how you travel — it watches that line



Between two plans

“+ Add route” → transit search → a travel-plan card with departure and arrival.

For any single plan

Bus · subway · train · plane · car, plus line, origin and departure time.

Then it watches

A strike on your line or a closure on your road becomes an alert for that plan.

Ask in your own words — answers come from real data only



1 **Code reads the question**

date · city · category · free · deadline · which plan

2 **Code picks real candidates**

up to 24 events from today's data — the AI sees nothing else

3 **Gemini answers inside them**

any id or number outside the candidates is dropped

4 **It proposes, you confirm**

[Add] / [Cancel] — and off-topic questions are politely declined

A coworker, not an oracle: AI writes, code checks



Facts

Public APIs

Dates, places, prices and links stay exactly as published.



Judgement

Rule engine

Which plan meets which event, distance, time window, score — deterministic and tested.



Language

Gemini

Explains, translates, reads news, understands your self-intro and questions.



Decision

You

Nothing is added, applied or saved until you tap.



Guardrails in code: every number in an AI sentence must exist in the input · ids must be real candidates · news quotes must appear in the article · self-intro evidence is cut from your own words by word index

We measured where AI fails — then changed the design

6 / 6

Card sentences passed the number check

Found: the model added an unverified “may affect your route” → crowd and traffic cards no longer use AI text.

6 / 150

News articles blocked

Strike reader: quote not found in the article → dropped instead of shown.

0

Invented evidence in self-intro

First design: 20–30% broken Korean quotes. Now the model points to word numbers and code cuts the text.

7 / 7

Assistant answers grounded

Zero events outside the candidates; off-topic questions declined.

Real Gemini API, team-built test sets (6 cards · 150 + 25 articles · 7 intros × 3 runs · 7 questions). Small-scale, not a labelled benchmark.

Three repositories, one pipeline



BE · data pipeline

GitHub Actions

- 3 public event APIs + NAVER news
- Normalize, validate, de-duplicate
- Gemini: translate, tag, read strikes
- Publish events · news · traffic JSON
- Daily 06:00, traffic every 3 h



AI · liferadar_core

Dart package

- InsightEngine · RouteMatcher
- TrafficMatcher · Taste
- InsightWriter · Assistant
- ProfileExtractor (+ validators)
- 105 tests, shared by FE and BE



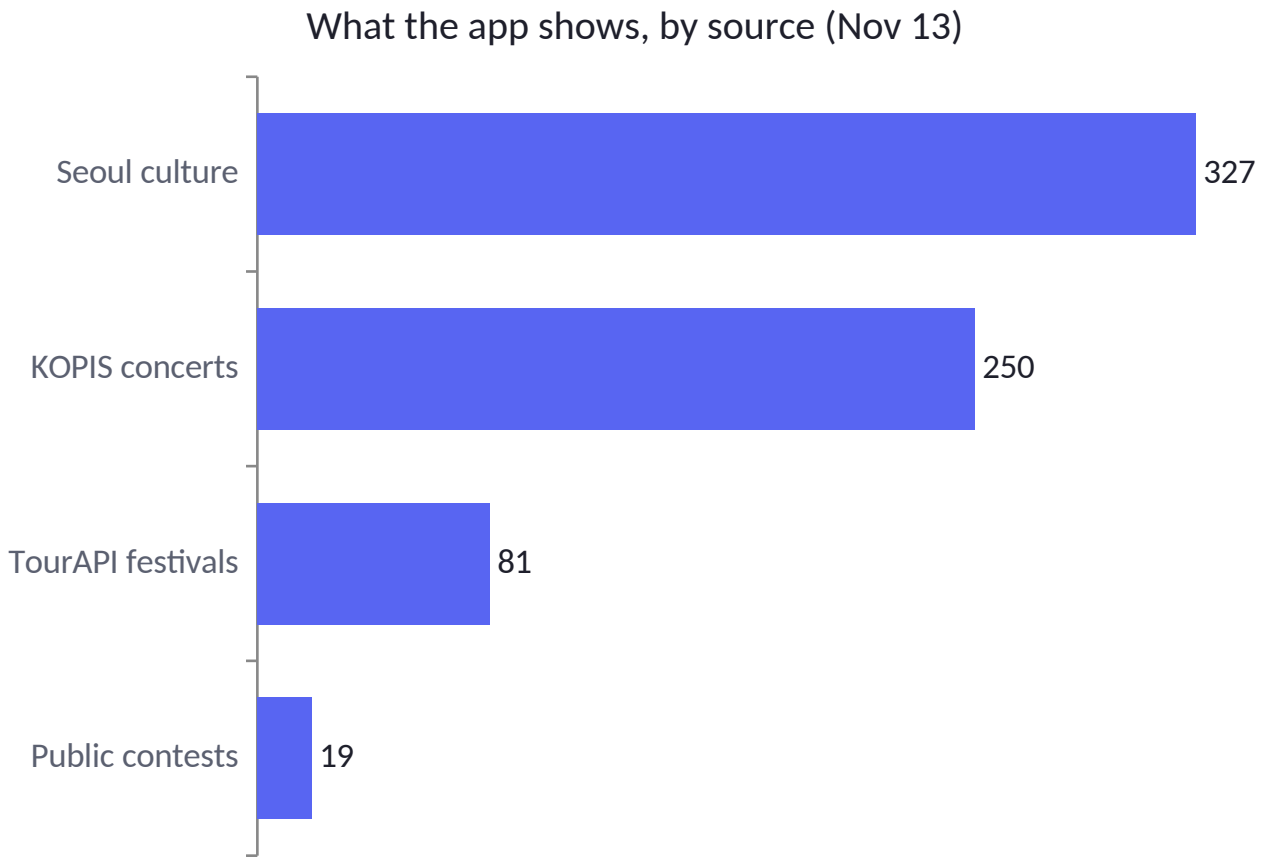
FE · Flutter app

Android + Web

- Cache-first feeds, works offline
- Google Calendar · Routes · Weather
- Kakao Maps · Local · Mobility
- National ITS queried on device
- Optimistic UI, ko / en, dark mode

Why ITS on the device? It blocks overseas IPs such as GitHub Actions — so the app asks it directly.

Real data, refreshed automatically



658 real events from 3 public agencies + 19 contests

292 → 11 news articles → transit strikes (AI-read, quote-checked)

26 official traffic notices merged (TOPIS, ITS, news)

Weekly contest collection from 3 public portals

Built to feel fast and stay up on a real phone

301

automated tests

FE 121 · AI 105 · BE 75 — CI on every pull request in all 3 repos

0.4 s

batching window

Weather and traffic results are batched — the app no longer froze for seconds after a plan change

20 s

network time limit

No endless spinners; friendly error toasts instead

0 ms

wait to see your tap

Like, add, save and delete update the screen first; failures roll back with an error toast

- Korean / English everywhere, event titles translated
- Light & dark themes, contrast ≥ 4.5 : 1, reduced-motion support
- Offline-first: last feed cached on the phone
- Real-device feedback → fixed (alerts tab, toasts, duplicate adds, freezes)

What is not done yet — stated openly

Sample data

IT conferences & hackathons have no public API → sample cards, marked “Sample”.

Push while closed

Alerts work while the app runs; server push (FCM) is designed, not built.

API keys in the app

For the demo; next: a server proxy and restricted keys.

City-level closures

Police UTIC key issued, waiting for server IP registration.

Flights

Recorded in plans, not used for alerts — no flight-status data yet.

Evaluation size

Small team-built test sets, not a labelled benchmark.

Next: server push · API proxy · UTIC city closures · flight status · university & international-student office pilots

AI finds.
You decide.

LifeRadar — a coworker that watches your plans, explains its reasons, and leaves the choice to you.

Install (APK)

[Insert Google Drive link + QR]

Demo video

[Insert YouTube link + QR]

Code

github.com/woowangthon — WOOWANGTHON_TEAM_27_FE · BE · AI

Notion

[Insert public Notion page link]

Team 27 DALAB · Abbos Aliboev · Jeon Yeongwoo

Questions we expect

Why not just use ChatGPT?

It waits for a question and can invent events. LifeRadar starts from your calendar, uses only today's real data, and code rejects unverified facts.

What if Gemini is down or out of quota?

Everything still works: rule sentences, filter search and alerts. AI text is marked "AI"; rule results are marked "filter search".

How do you stop hallucinations?

Code retrieves facts first; numbers, ids and quotes must exist in the input or the sentence is replaced.

What about privacy?

Interests, own plans and feedback stay on the phone; the calendar token stays in memory; chats are not stored.

Is a crowd warning reliable?

We never claim traffic impact from an event alone — it is a "reference" card until an official notice exists.

How would this grow?

API proxy + server push, then pilots with universities and international-student offices; data is national already.

Sources, what we built, and who built it

Data / external tech	Used for
Seoul Open Data · KOPIS · TourAPI	Events, concerts, festivals
Public contest portals (Sotong24, Linkareer …)	Contests and deadlines
Seoul TOPIS · national ITS · (Police UTIC pending)	Official road closures
NAVER News API → Gemini	Transit strike reading
Google Calendar · Routes · Weather · Gemini · Sign-In	Plans, transit routes, forecast, AI text
Kakao Maps · Local · Mobility	Map, place search, driving routes
Flutter, Riverpod, go_router (open source)	App framework

Public data under the Korea Open Government License (KOGL); sources are shown on every card.

Built by us

All app UI and state, the rule engine, AI prompts and validators, collectors and schedules, design system, tests. An AI coding assistant (Claude) was used and is credited in commits.

Team 27 DALAB

Abbos Aliboev
Planning, architecture, design; app, pipeline, engine & AI

Jeon Yeongwoo
Traffic research & impact rules; plan routes, traffic v2, contest collection